

prostep ivip
**SYM
POSIUM
2025**



T Systems



PDM Interoperability Forum

A Spotlight on Fasteners Impact on Innovation at BMW

REVOLUTIONIZING INDUSTRIES WITH DIGITAL TWINS AND AI

BCC BERLIN
13-14 May 2025

www.prostep-ivip-symposium.org

Florian Rank | Guillaume Hirel

Premium Partner



<https://www.mbx-if.org/home/pdm/>



Interoperability Forum

Key Takeaways



What is PDM-IF?



What are the Fastener Use Cases of BMW?



Further PDM-IF Fastener Use Cases and Outlook for the use of STEP AP242 XML and xMCF

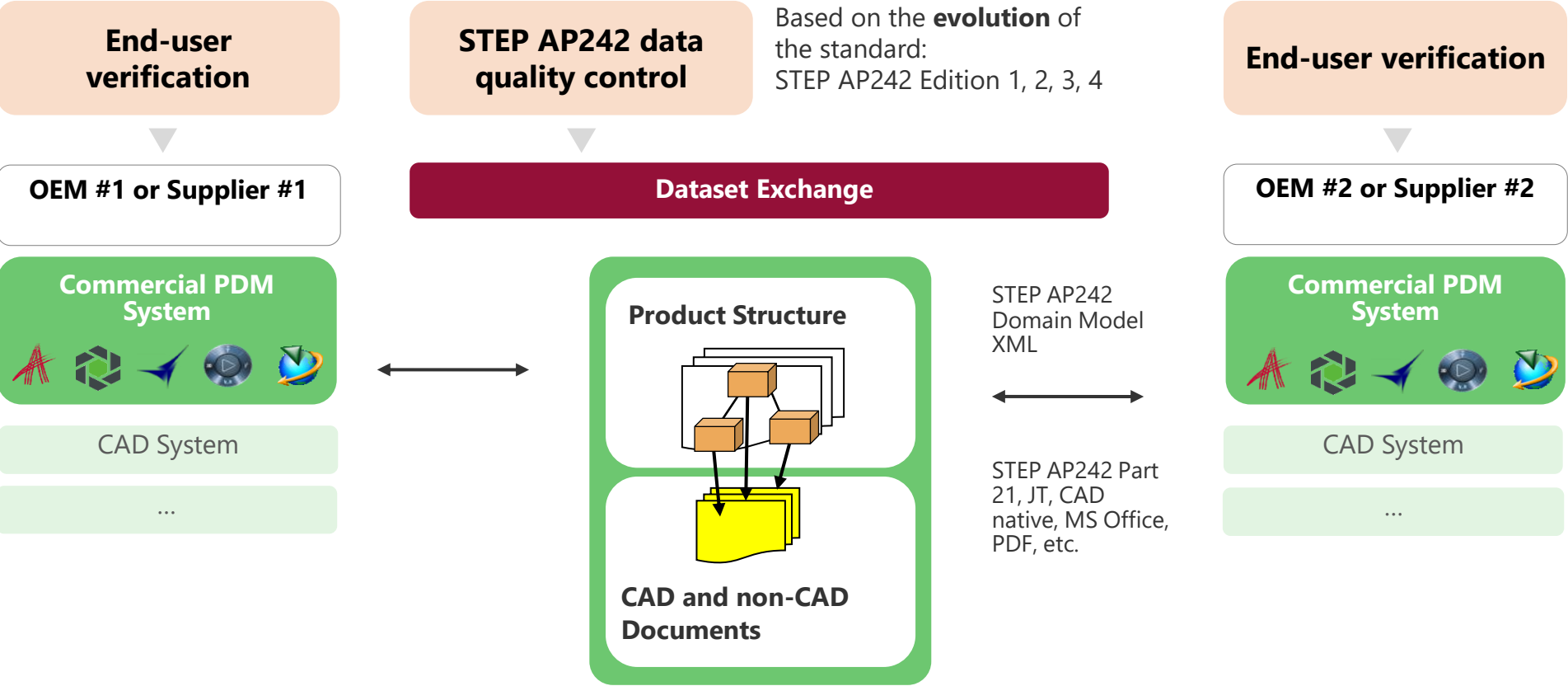




WHAT IS PDM-IF?

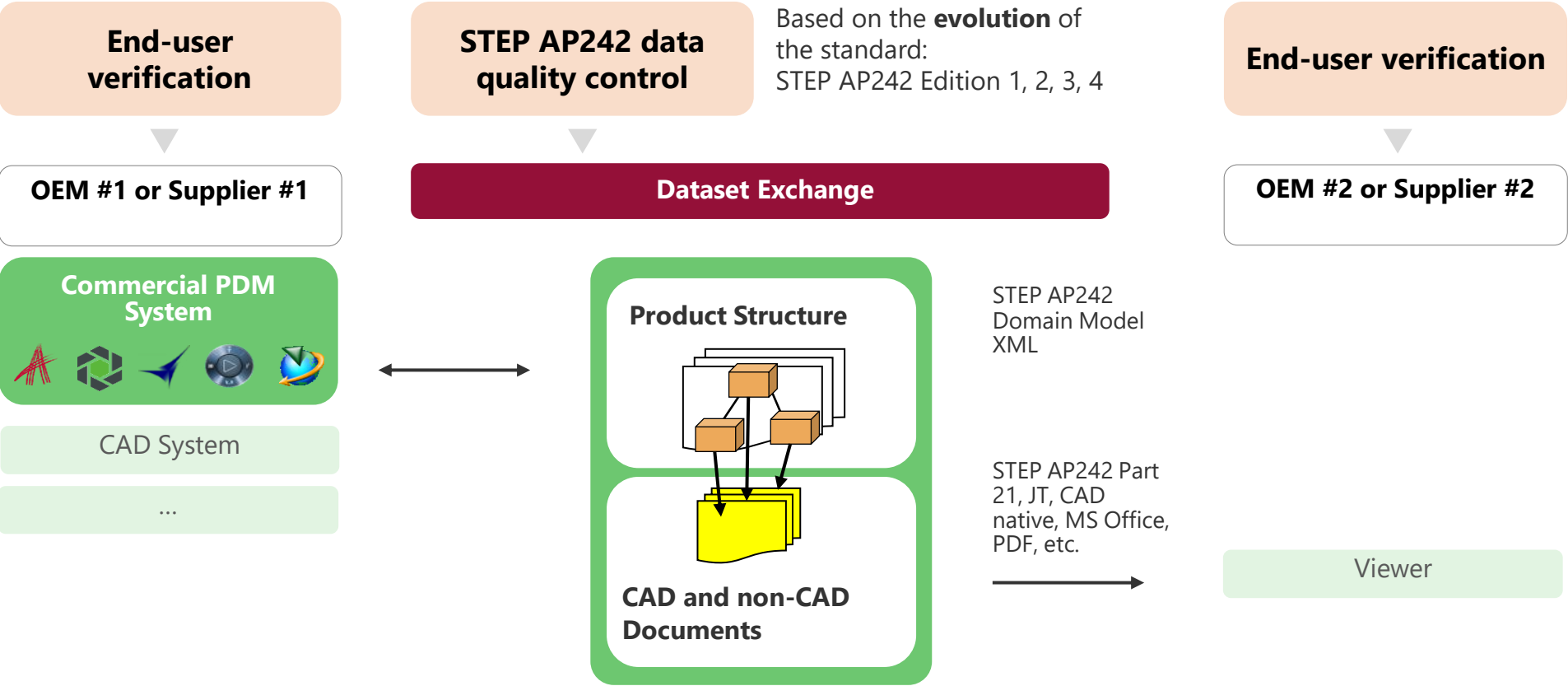
General PDM-IF Use Cases

PDM<->PDM or PDM<->CAD Data Exchange



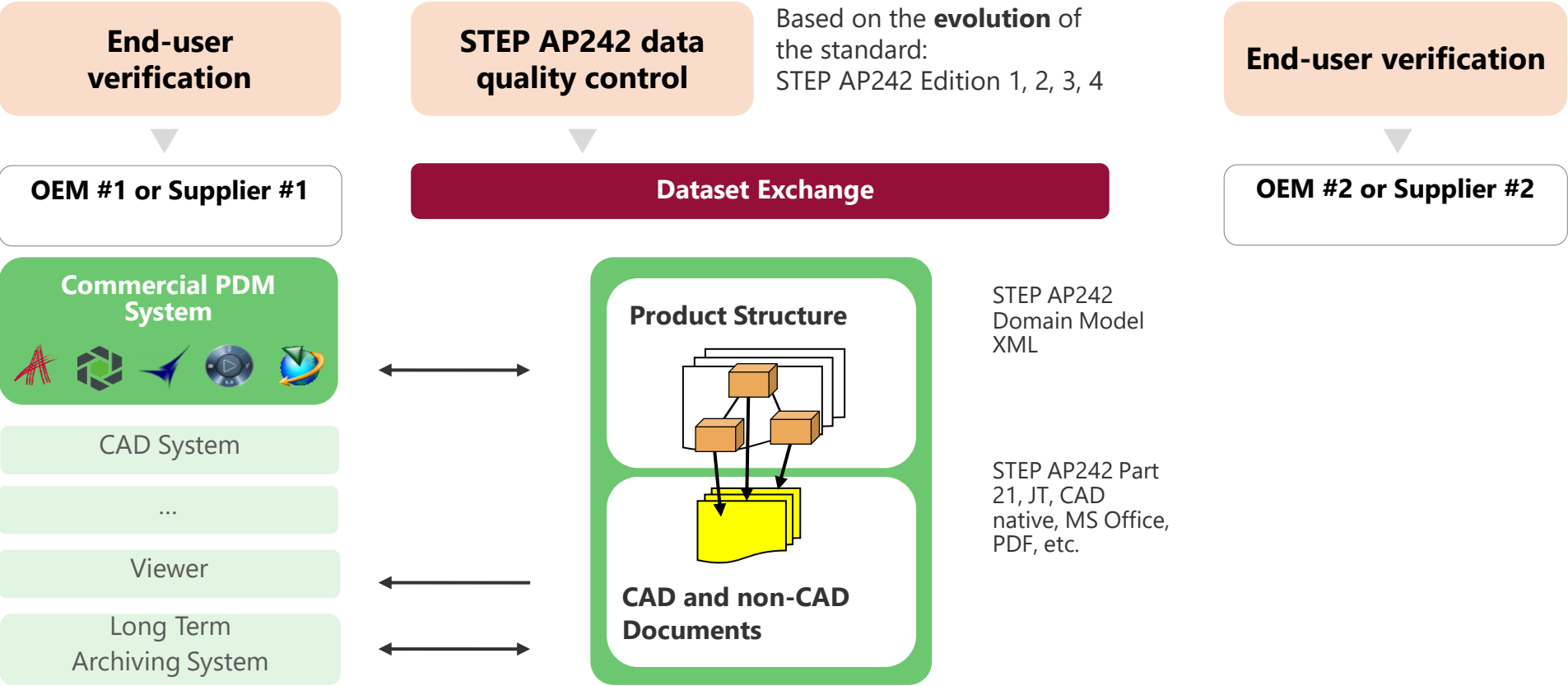
General PDM-IF Use Cases

PDM<->Viewer Data Exchange

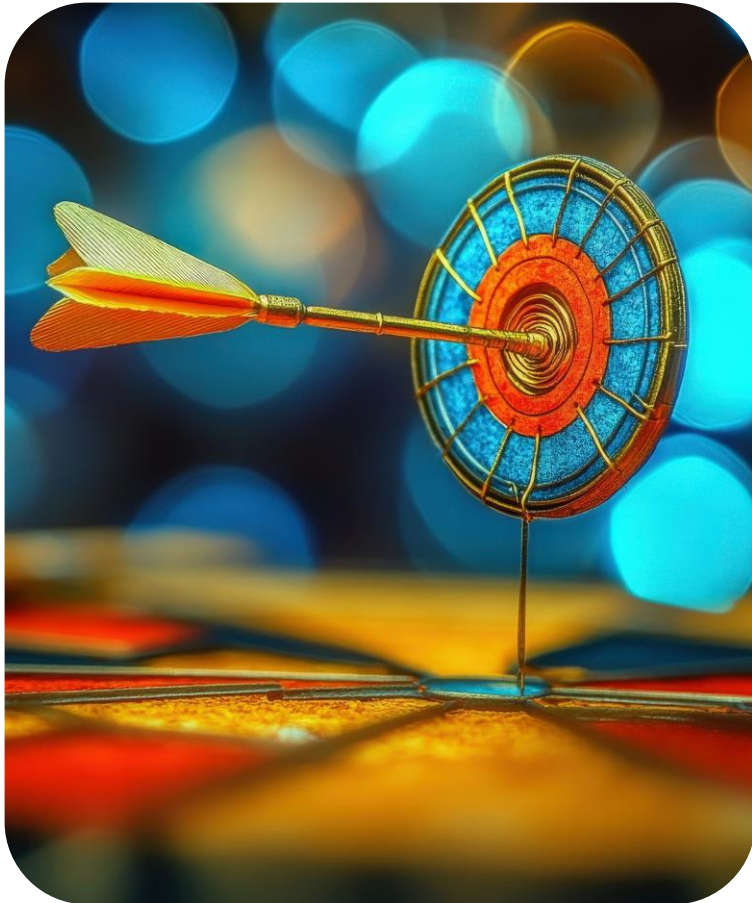


General PDM-IF Use Cases

PDM<->Archiving or Viewing out of Archive



Goals



- Provide validated Recommended Practices for industrial use cases, guiding through the capabilities of the standard => "STEP Cookbook"



- Enable vendors to shorten development cycles and increase the stability / compatibility of their STEP interfaces



- Ensure stability of new capabilities and support longevity of data, hence safeguarding investments



Project Members

Associations

**Industrial
Project Leaders**



Jérôme Brun (BMW)

Coordinators

Guillaume Hirel (prostep ivip)



Frédéric Darré (Airbus)

Kevin Le Tutour (AFNeT)

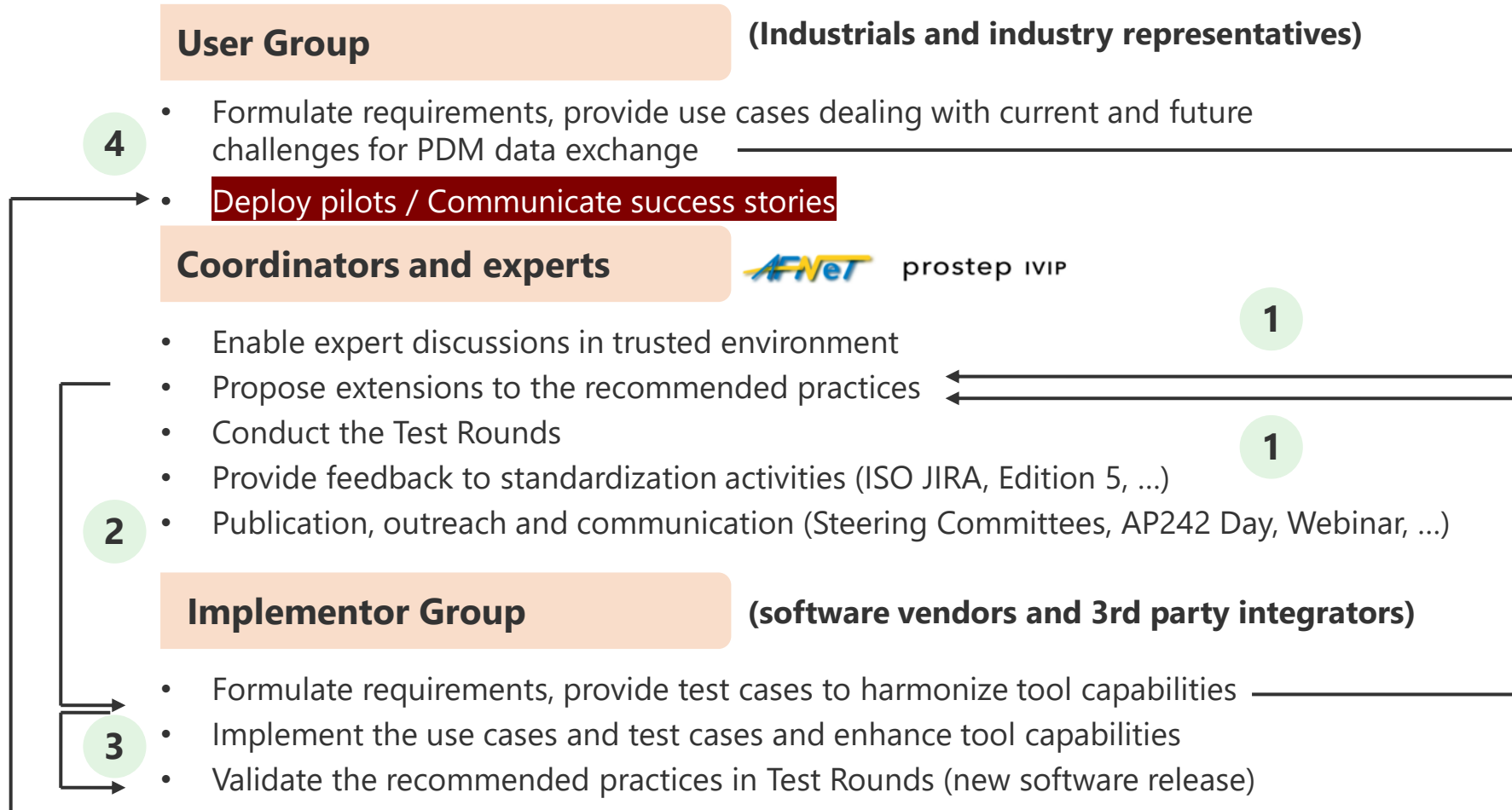
User Group



Implementor Group



Roles of User Group and Implementor Group



STEP AP242 XML Domain Model

A cross-domain Backbone

- PMI
- B-Rep & Tessellated Geometry
- User Defined attributes
- Validation properties...



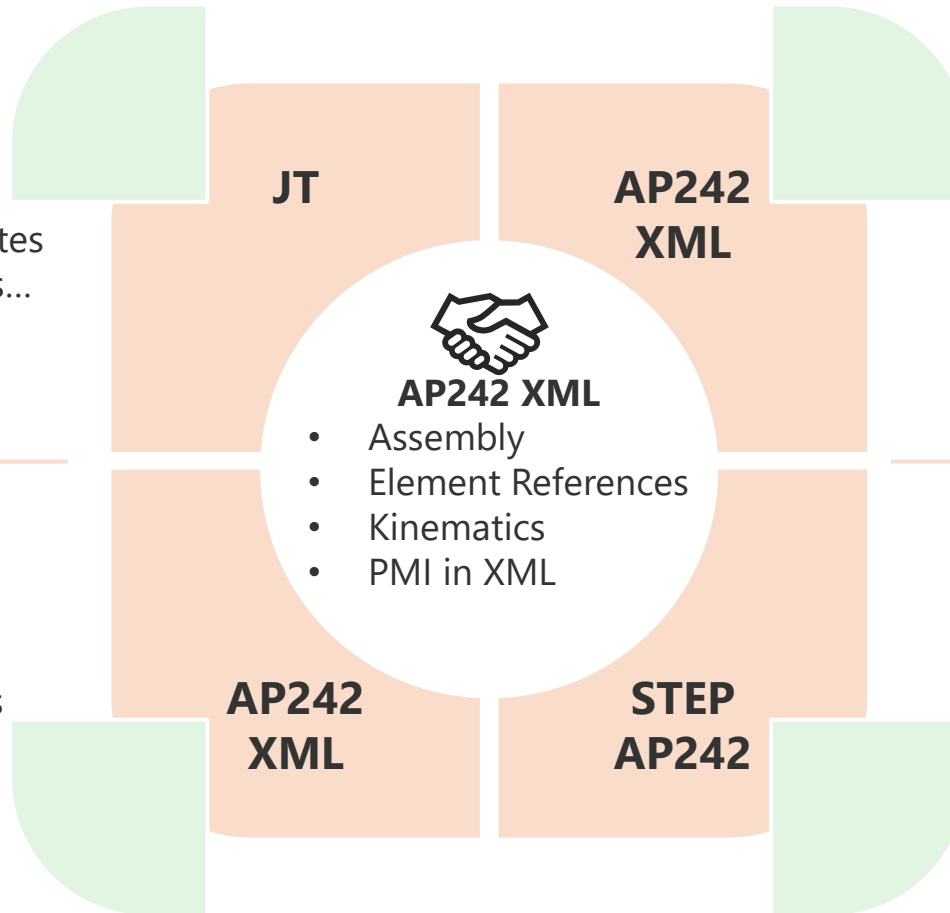
- Configuration
- Change Mgmt.
- Visual Issue Mgmt.
- Validation Properties



- Harness Topology
- Connectivity...

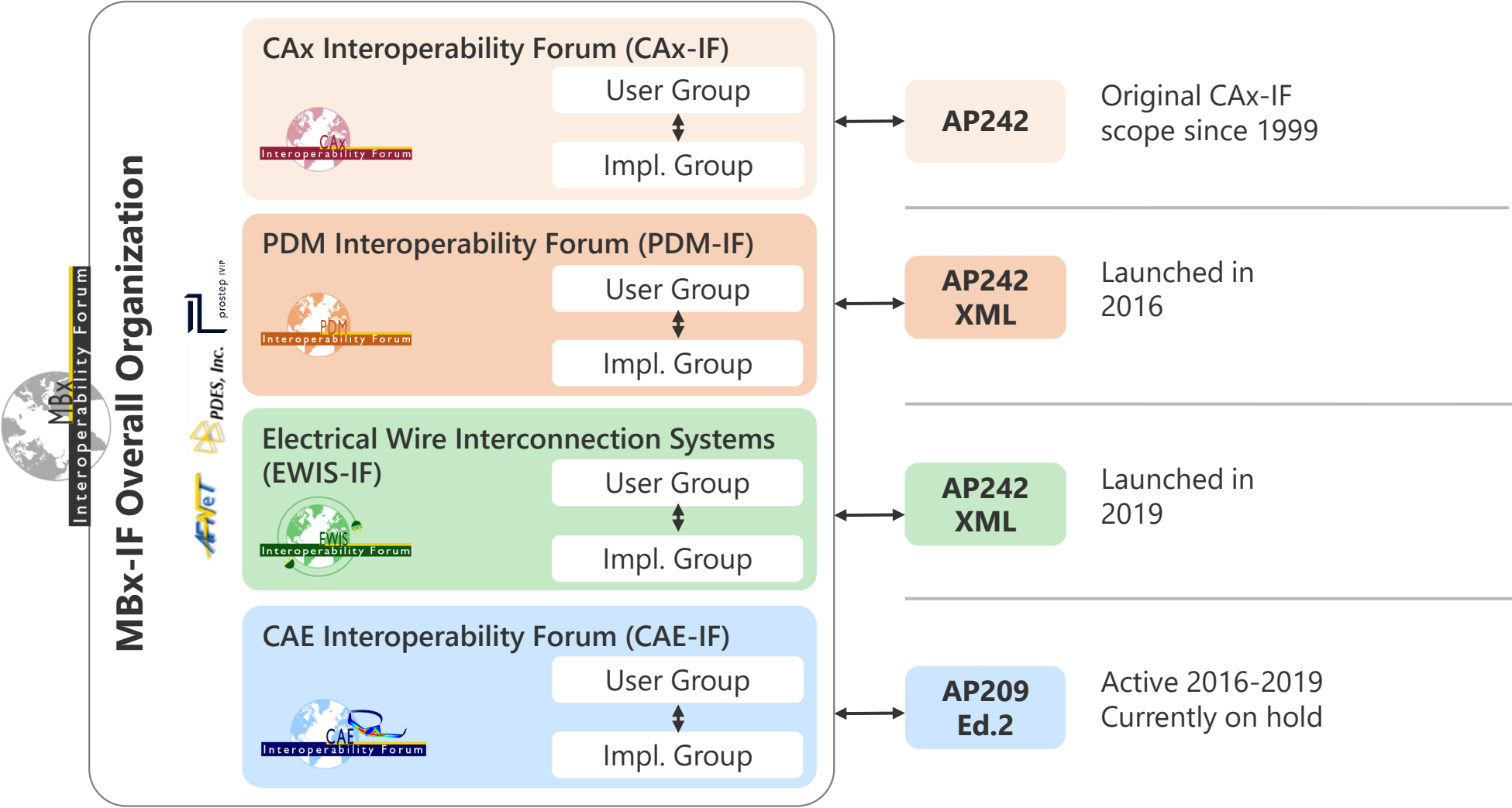


- PMI
- Composites
- B-Rep & Tessellated Geometry
- Assembly & Installation
- User Defined Attributes
- Validation Properties...



MBx-IF

Joining the efforts



PDM-IF Test Rounds Summary

BM=Benchmark

20242025

PDM-IF UC/TR Matrix	1-9	10	11	12	13	14	15	16	17	18	19	20
Assembly Exchange with Org. Data and product structure changes	1-3 + BM#2 Done											
Exchange of 150% BoM (simple and complex effectivities)	5-7 Done											
Visual Issue Management (VIM)		Preparation							Planned Ed4		UC1 Planned	
As designed / As planned (multi-BoM)	9 Started								Restarted	Done		
RV&V (Promethea)		Preparation					Stopped (SE-IF)					
Multiple Id & Customized PDM Properties	8 Started		BM#3 Done									
Configuration Management UC1/2/3		UC1 UC2 Preparation	UC1 Done	UC2 Done	UC2 Nested Done	UC3 Preparation		BM#4 UC 2 Done			UC3 On hold	
Alternate and Substitute Parts				Started	Nested Done			BM#4 Done				
Incremental exchange				Started	Nested Done							
Nested file exchange	4-6 + BM #3 Done				Updated				Planned Ed4 + Part15 Ed2	Started		
Mirrored Parts						Done						
Alt. Geometries and User Defined Attributes							Done					
Product structure validation properties (PSVP)					Preparation			Planned		LOTAR Pilot Ed4	On hold	
Version Branch Effectivity							Preparation		Preparation Ed4	Started		Planned
Flexible Parts							Preparation		Started	Done		
Delta Exchange							Preparation		Started		Planned	
Property Groups							Preparation	Started		Done		
Work Order for Design Change								Preparation			Planned	
Export control: Data Classification and Export Licenses								Preparation				Planned
Fasteners								Preparation				Planned
Compression (stpXZ) and Archive (stpA)										Started Done		
Re-import enhanced performance									Preparation	Started		
Material Properties									Preparation			Planned





FASTENERS USE CASES @ BMW



Using χ MCF in the BMW Car Development.



Development

Production



Geometrical Design and Data Management.



Joining Technology

In-house
IT-Solution as
data master



Geometrical Data Management

In-house
database
as data master



Geometrical Design



Development

Production



xMCF Format for Joining Technology Data is used for Visualization and data transfer.



Joining Technology

In-house
IT-Solution as
data master



Geometrical Data Management

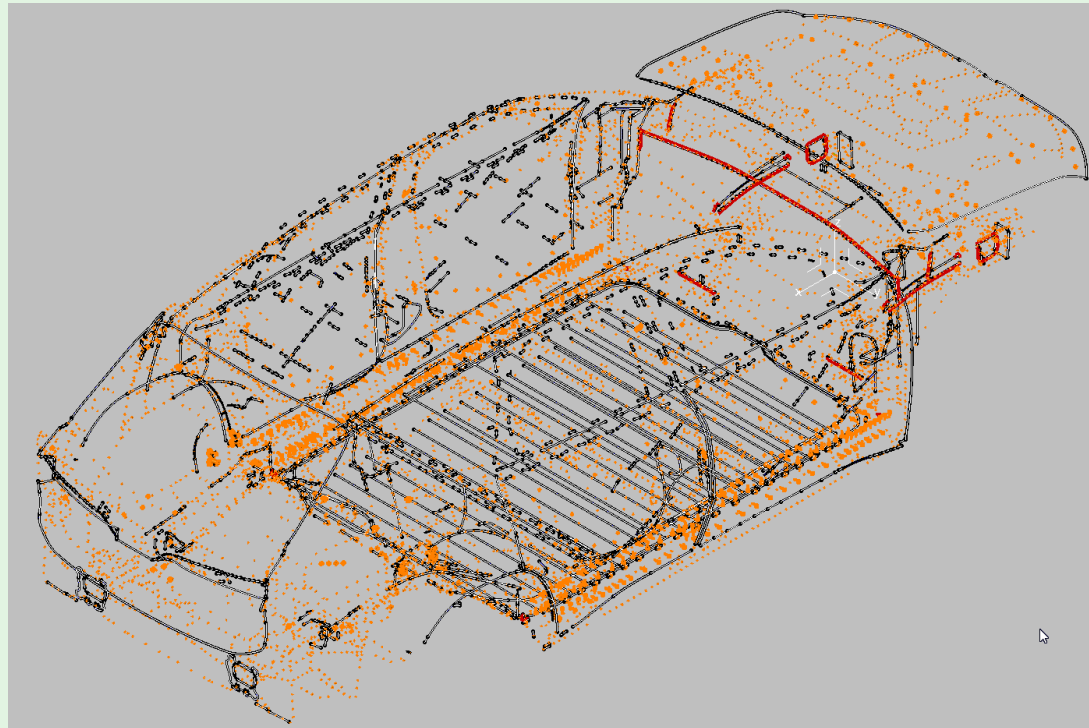
In-house
database
as data master



Geometrical Design



xMCF



Geometrical Data (incl. Metadata) are transported in different Formats.

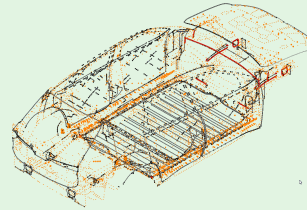


Joining Technology

In-house
IT-Solution as
data master



χMCF

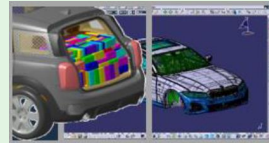


Geometrical Data Management

In-house
database
as data master



CATPart, PLMXML, ...



Geometrical Design



Use Case Geometrical Data for Preprocessing ...

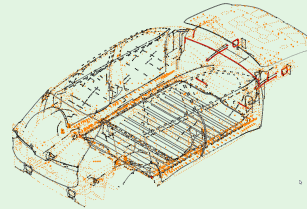


Joining Technology

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IT-Solution as
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χMCF



Geometrical Data Management

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CATPart, PLMXML, ...

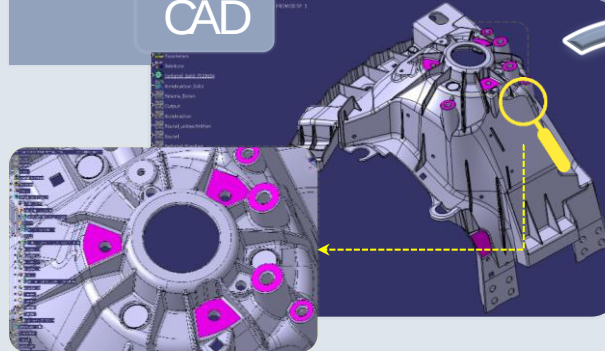


Geometrical Design



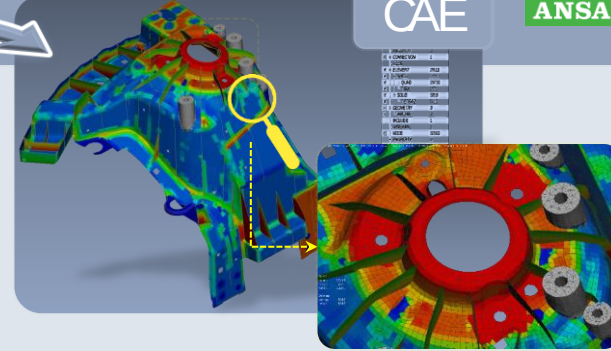
CAE-Prozesskette / FE-Modellaufbau

CAD



CAE

ANSA



... and Simulation.

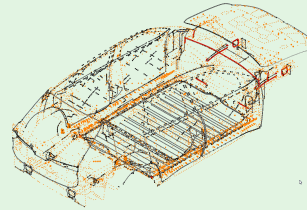


Joining Technology

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χMCF

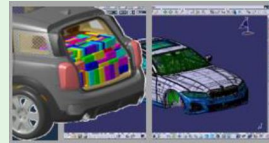


Geometrical Data Management

In-house
database
as data master



CATPart, PLMXML, ...



Geometrical Design



Simulation / Auslegung

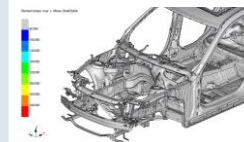
Crash



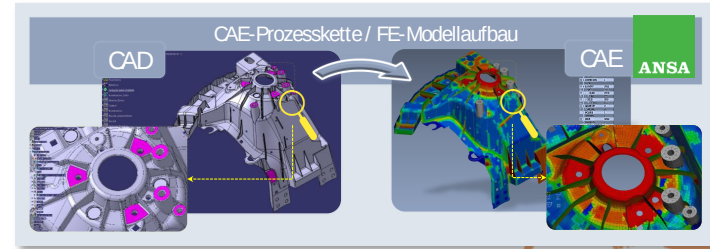
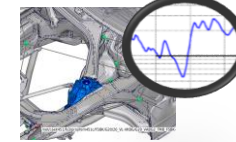
Dynamics



Endurance

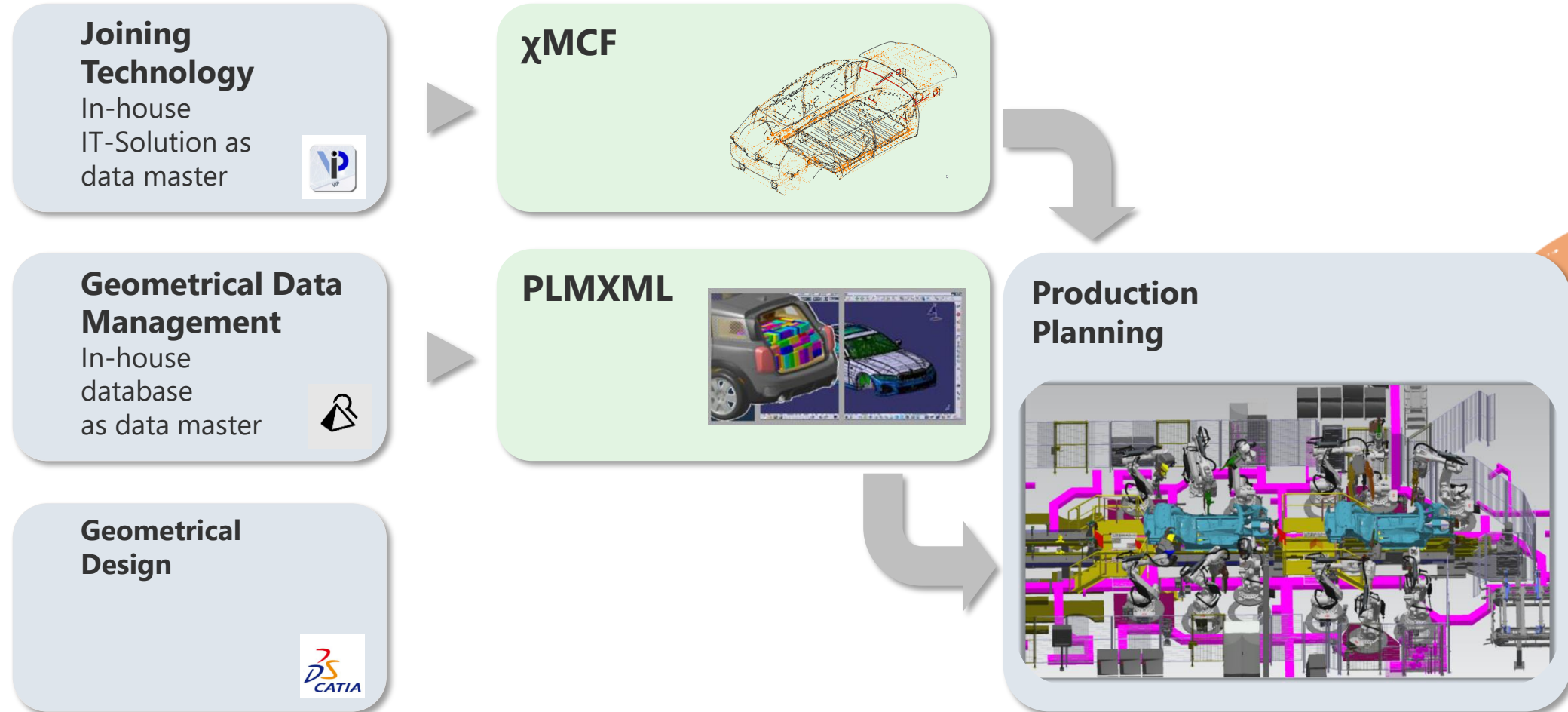


Acoustics



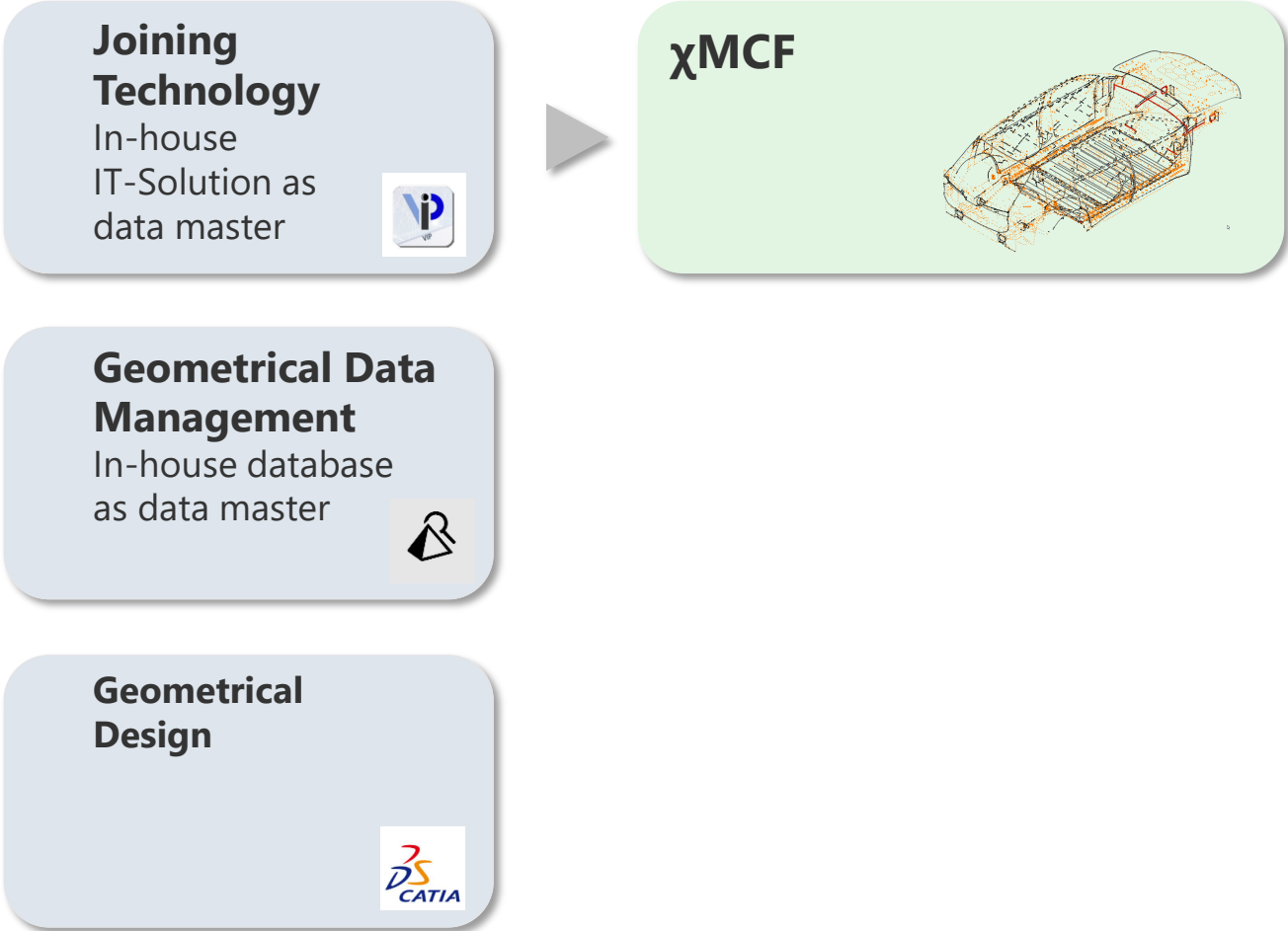


Use Case Geometrical Data for Production Planning.

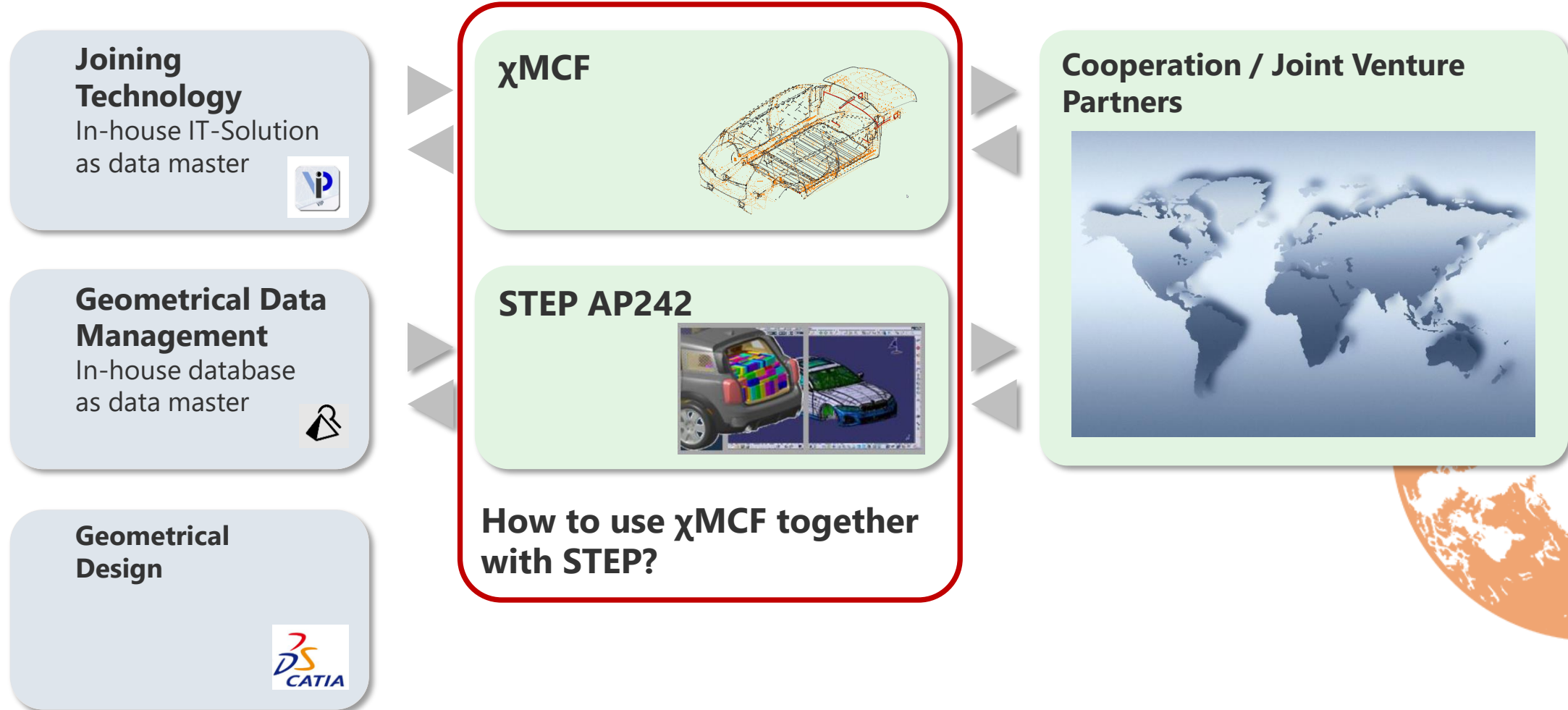


χMCF is already used for Some Scenarios at BMW.

The Inhouse Solution for Joining technology will Remain as it is long term.



For exchange of Geometrical Product data for example with partners using different PLM Systems, STEP AP242 is a very valuable Format.





FASTENERS USE CASE @ PDM-IF



PDM-IF Scope of Fasteners

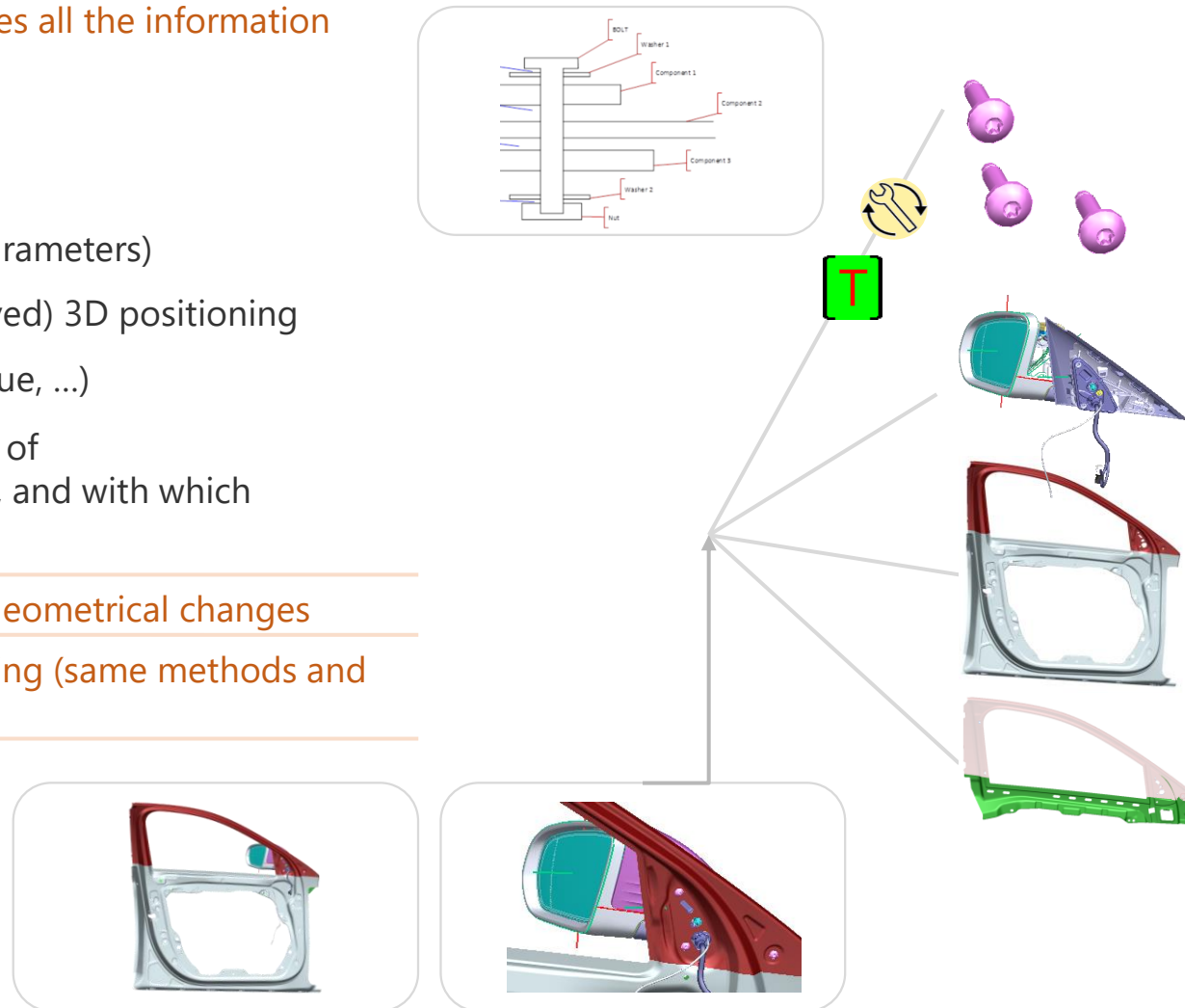
Mating bundle object, that holds/structures all the information about:

- Which parts are mated together
- In which order (stacking)
- With which fasteners (defined by parameters)
- Direction of the fasteners and (derived) 3D positioning
- Mating parameters (tightening torque, ...)
- In case of 150% BoM: configuration of which fasteners are involved, where, and with which parameters

Stable Id of the Mating object in case of geometrical changes

Used in the same way for all types of mating (same methods and functions, with different parameters)

Additionally in scope for CAx-IF and EWIS-IF: geometric elements (geometric constraints, contact faces, ...)



PDM-IF Scope of Fasteners

Further aspects



What is special about Fasteners?

- Function dominates over their shape
- CAD/BoM design and fasteners complement each other. Each is useless without the other
- Different expert knowledge, process steps and tools than CAD/BoM design
- Use cases: durability, simulation, robot programming, supplier integration,...

Modern PLM Features regarding Fasteners

- Propose concrete fastener parts out of the Fastener parametric definition, the thickness of the linked parts and the material constraints
- Compute (from material constraints) their number and positioning (from the wished orientation and for example equidistant)
- Parametric definition of the holes the linked parts (no physical design of the holes)

=> for an automated design, supporting changes in the product design with less effort

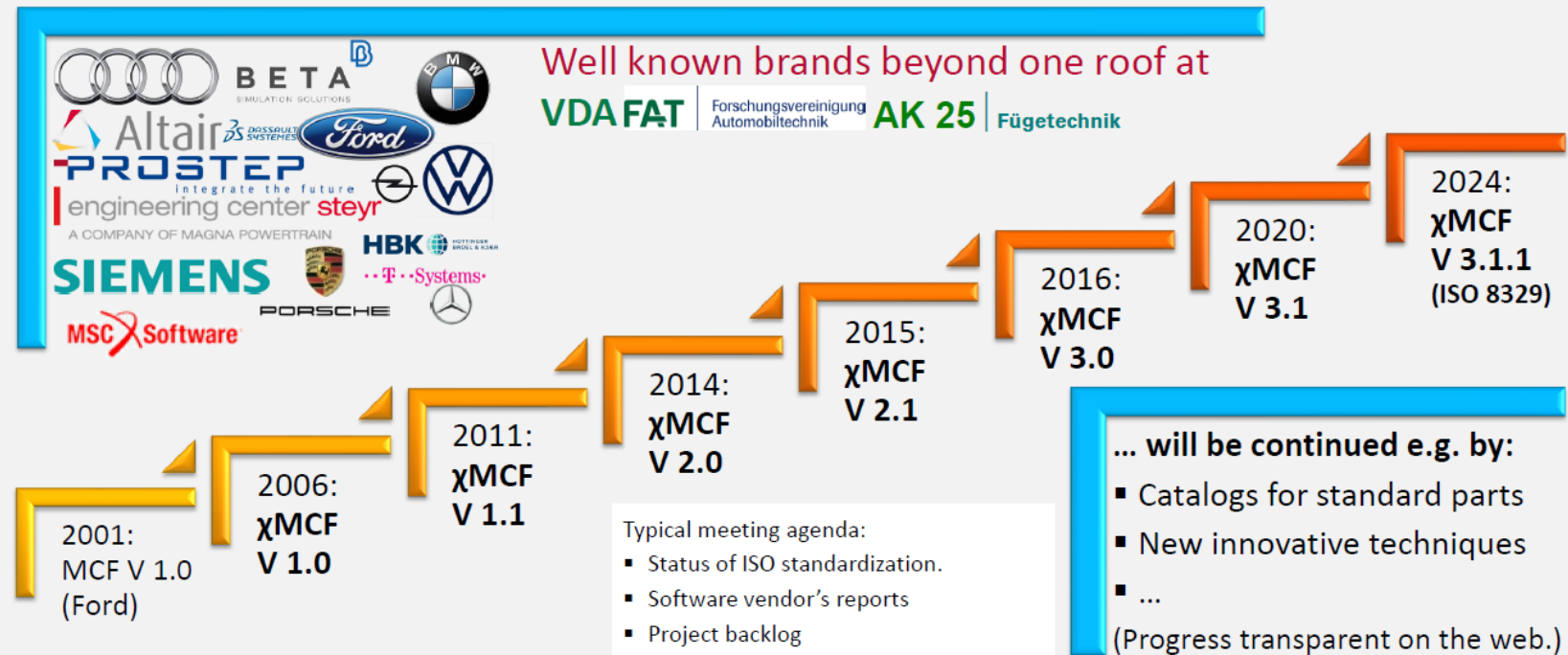


χMCF (ISO 8329 PAS)

Overview

Standard for the description of mechanical connections

> 60 connection techniques, up to 25 quality criteria per connection technique



Use Case Status @ BMW

Full dataset ready

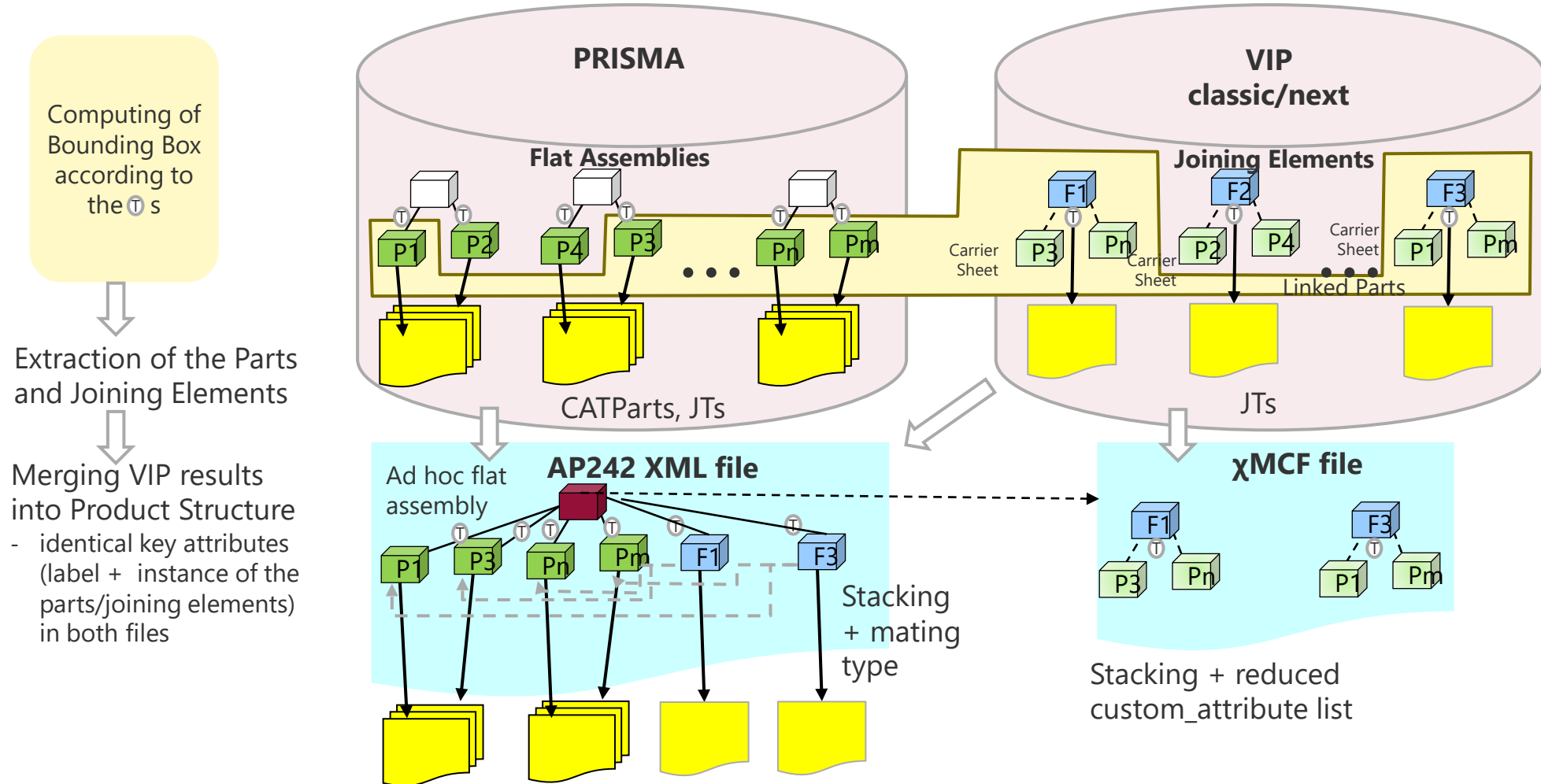
- Assembly Structure mapped from Prisma->CATIA V5->AP242 XML Edition 3 + P21/JT geometry
- Enriched with Joining Elements definition, and references to χ MCF file
- χ MCF File (from VIP) containing the structure and the reference/instance attributes of the Carrier Sheets and the Joining Elements
- Extensive use of customized χ MCF attributes
- Beta CAE could read this dataset into ANSA

Analysis of the used χ MCF custom_attributes

- Proposal to move some of them to AP242 XML and complete the dataset with combination of AP242 and χ MCF



Use Case Status @ BMW



Synergy within the PDM-IF User Group



Similar Use Case at further OEMs (aerospace)



Intensive cooperation to get a quick solution and short-term implementation

First proposal for AP242 Edition 5 that fits for the whole MBx-IF

IP Protection of each OEM methodology is kept



Outlook



PoC implementation using AP242 Edition 4 + χ MCF 3.1.1 from the IT vendors @ PDM-IF



Proposal for AP242 Edition 5

- Consolidated from PDM-IF, CAX-IF, EWIS-IF
- To be consolidated by further WGs



Issues to χ MCF 3.1.1

- Define User in charge of pushing the issues
- Add missing mating types
- Need to move from ISO PAS (Publicly Available Specification) to IS (International Standard) within the next years



PDM Interoperability Forum

Join the ride!



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Guillaume.Hirel@t-systems.com
(project coordination prostep ivip)

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